

TRANSCRIPT SUMMARIZER

Meeting Guidelines



AGENDA

Problem Statement

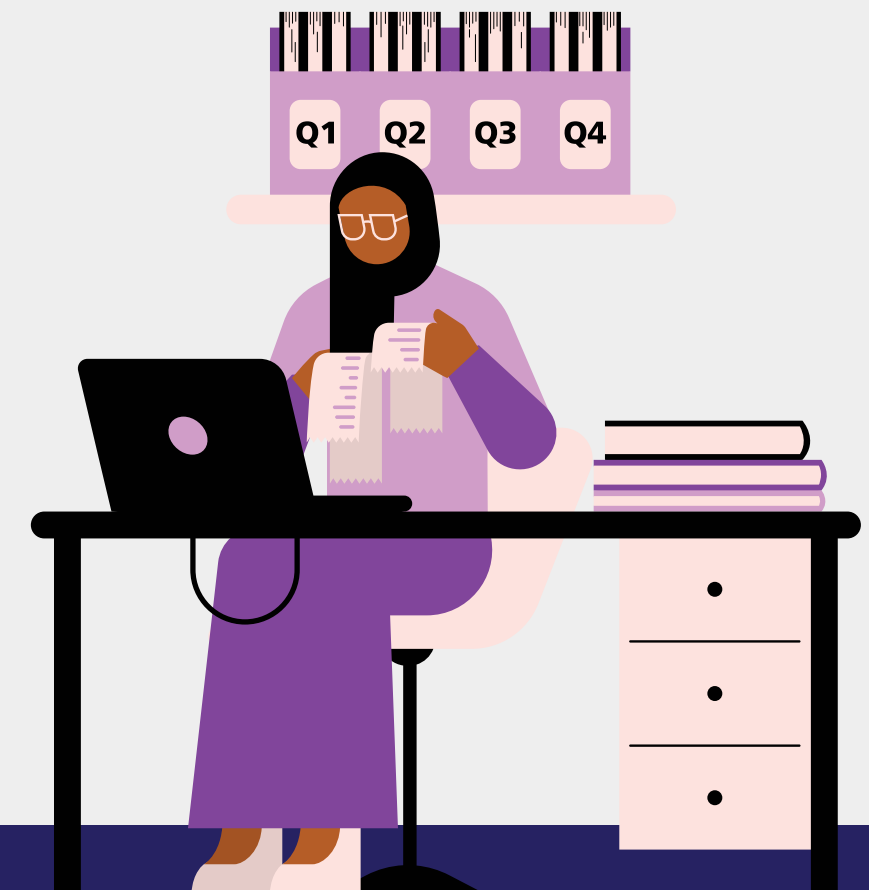
Dataset Overview

Fine-Tuning Process

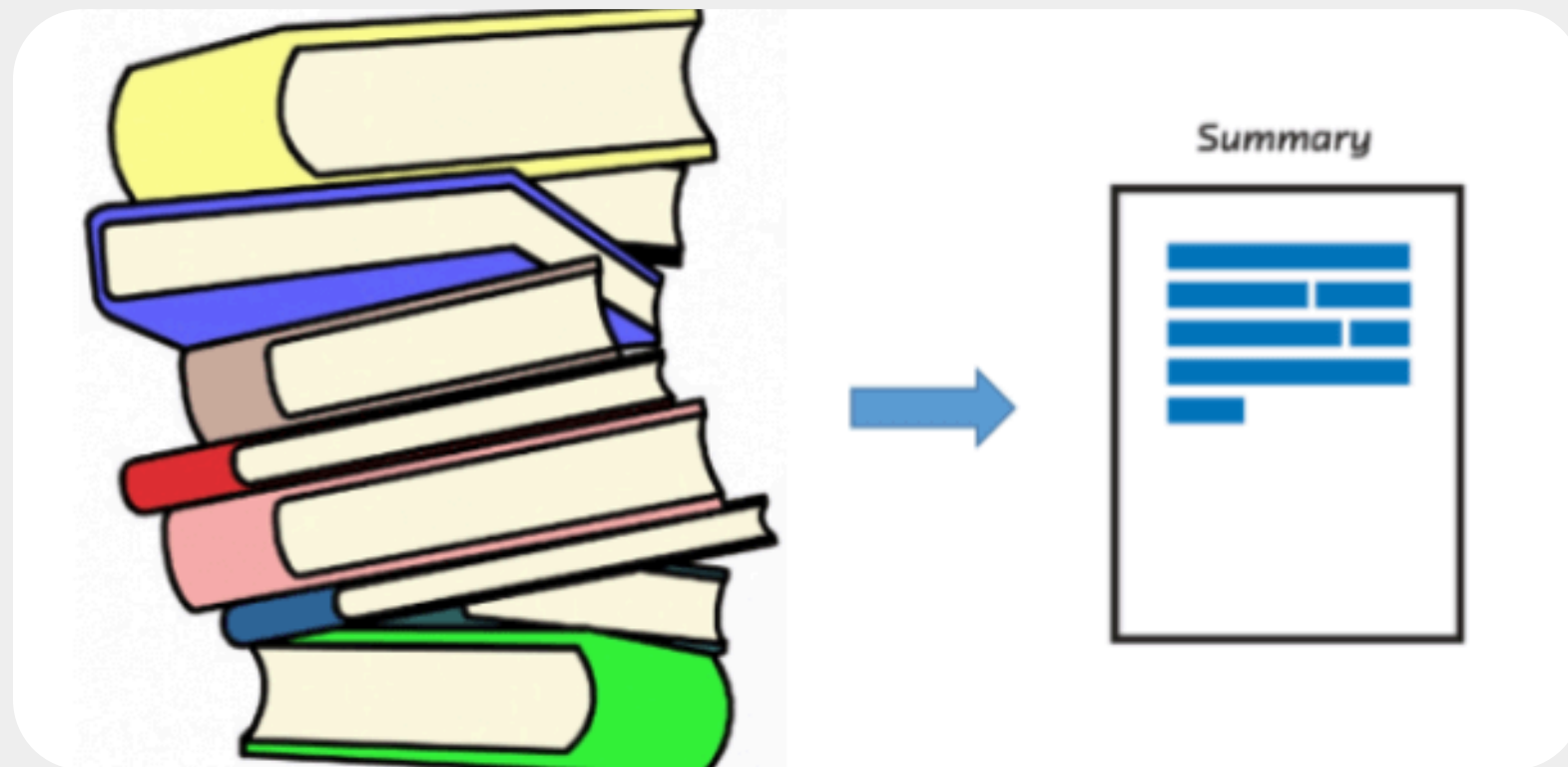
Results & Evaluation

Challenges &
Solutions

Demo



PROBLEM STATEMENT



Meetings generate long transcripts

Manual summaries are time-consuming & error-prone

Need for automated, accurate summarization

Why it matters:
"80% of meeting time is spent reviewing prior updates"



Zoom/Teams
recordings



Business weekly
sync-ups



Project
management
updates



Customer support logs

REAL-WORLD MOTIVATION

Challenges encountered during
project work

Key Benefits:

- >> Improved efficiency
- >> Knowledge retention
- >> Compliance





DATASET

Tokenization: Byte Pair Encoding to handle out-of-vocabulary words.

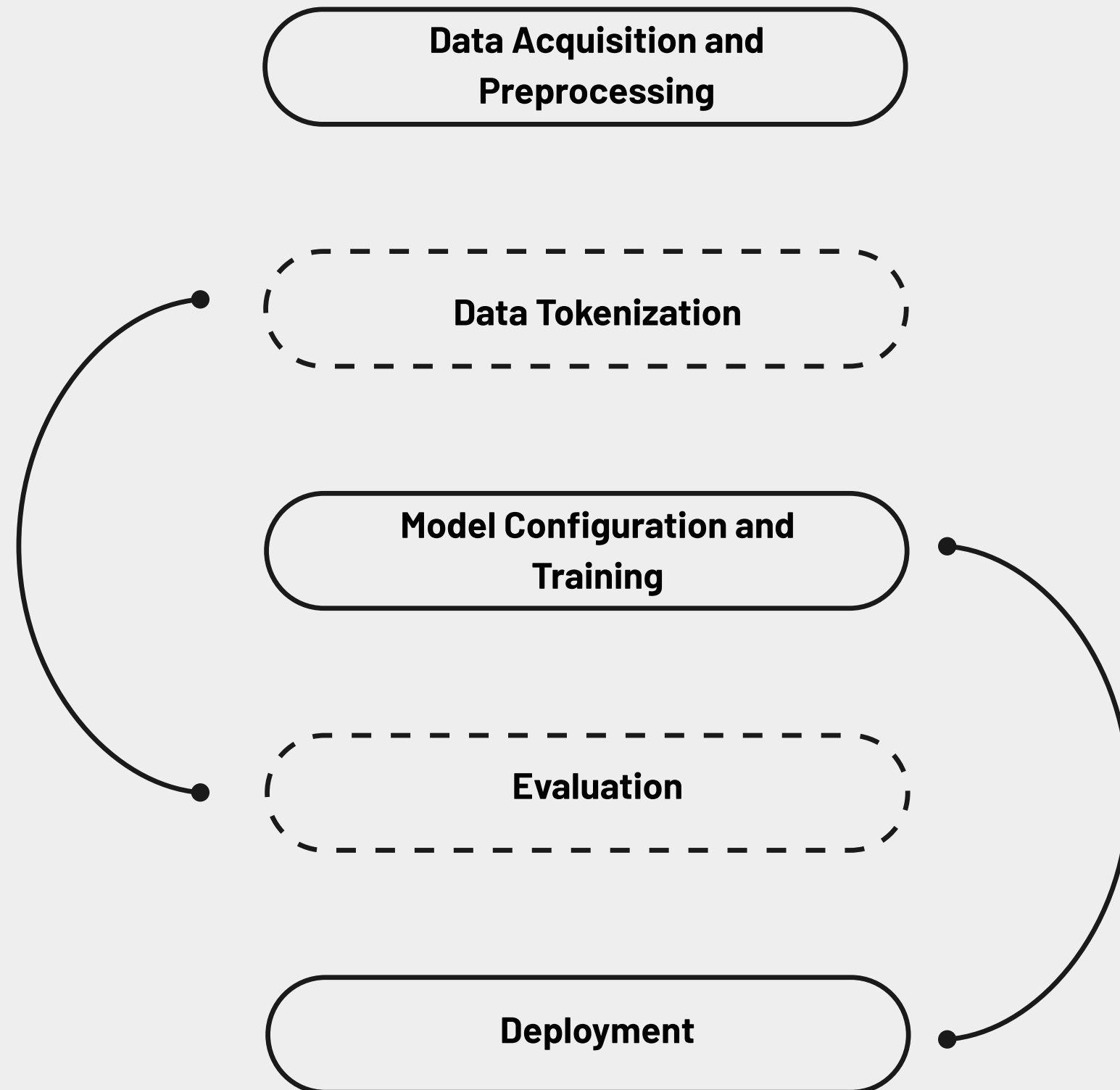
Text Cleaning: Remove special characters, numbers, and unnecessary white spaces. Normalize text (convert to lowercase, remove URLs, speaker labels/names).

Sequence Padding & Truncation: Ensure consistent input sequence lengths using padding and truncation.

Max Length: 512 tokens for transcripts
128 tokens for summaries

Detail	Compact	Column	6 of 6 columns			
id	Item_UID	Meeting_UID	Transcript	Summary	Date	
862 unique values	862 unique values	559 unique values	862 unique values	860 unique values	 2014-01-06 2022-08-09	
LongBeachCC_01072014_14-0024	14-0024	LongBeachCC_01072014	Speaker 2: And 14 is communication from the Office of the Council Member. I lost the recommendation ...	Recommendation to direct City Clerk to place an advisory question on the April 8, 2014 citywide elec...	2014-01-07	
DenverCityCouncil_01132014_13-0950	13-0950	DenverCityCouncil_01132014	Speaker 1: 12 eyes. 12 eyes. The bill is has passed. Now we come to our last public hearing. And Cou...	Adopts the 61st and Pena Station Area Plan as a supplement to the Comprehensive Plan. (LAND USE, TRA...	2014-01-13	
LongBeachCC_02042014_14-0092	14-0092	LongBeachCC_02042014	Speaker 4: There are conditions on this one. This is recommendation to receive an f, receive the app...	Recommendation to receive the application of Abdul Jamal Sheriff, dba Come N Go Market, for an origi...	2014-02-04	
LongBeachCC_02112014_14-0110	14-0110	LongBeachCC_02112014	Speaker 5: Thank you. We'll go to item six. Quick read. Speaker 6: Item six is a report from the off...	Recommendation to request City Manager to direct Airport Director to provide a report on how the Cit...	2014-02-11	

PROJECT WORKFLOW OVERVIEW





T5 TRANSFORMER



TEXT-TO-TEXT TRANSFER TRANSFORMER

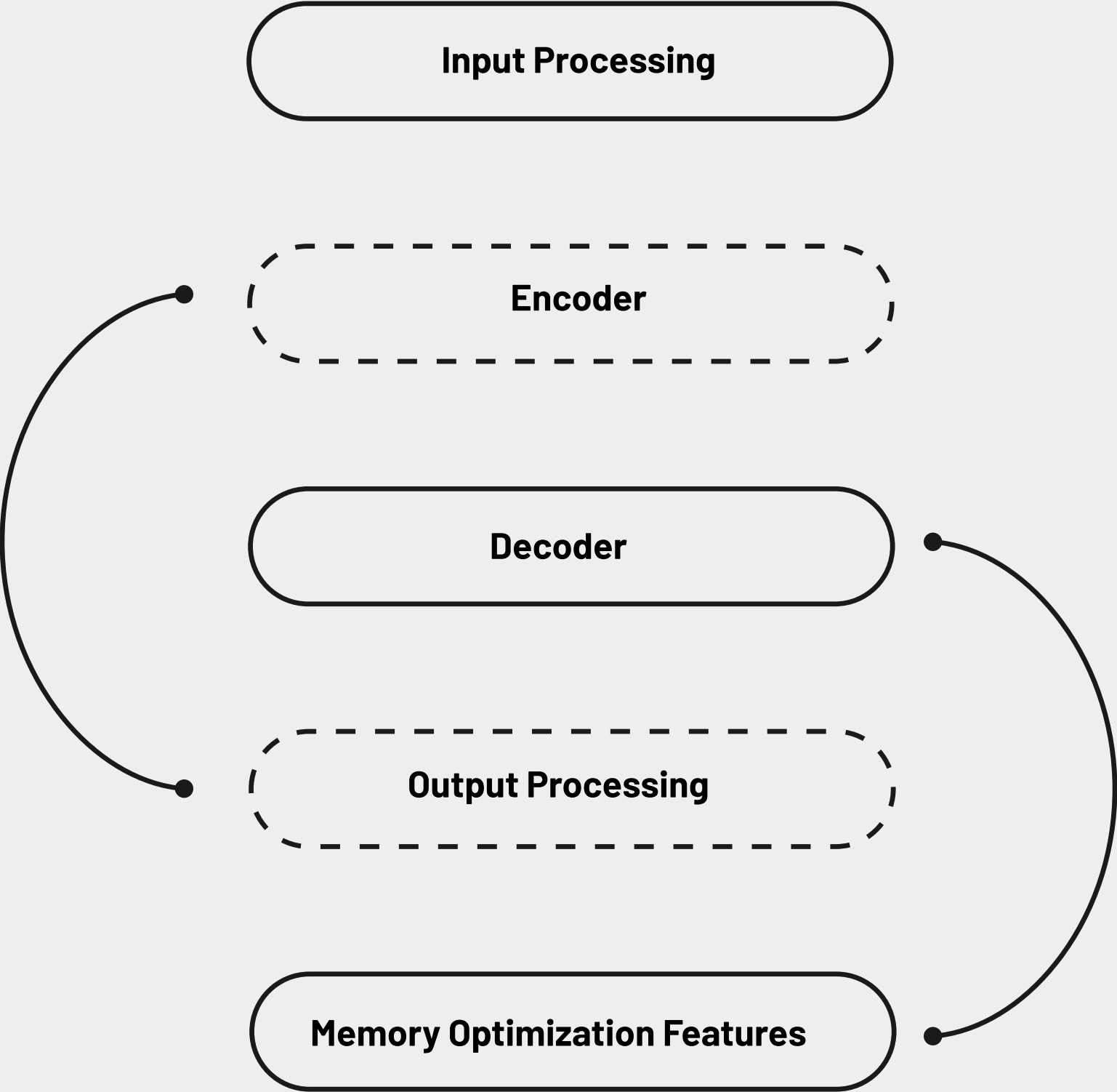
✓ Treats all NLP tasks as text-to-text

✓ Easy to fine-tune by simply changing task-specific prompts.

✓ Pretrained on a massive dataset (C4)

✓ Well-suited for abstractive summarization, especially on long, messy text like meeting transcripts.

T5 MODEL ARCHITECTURE OVERVIEW

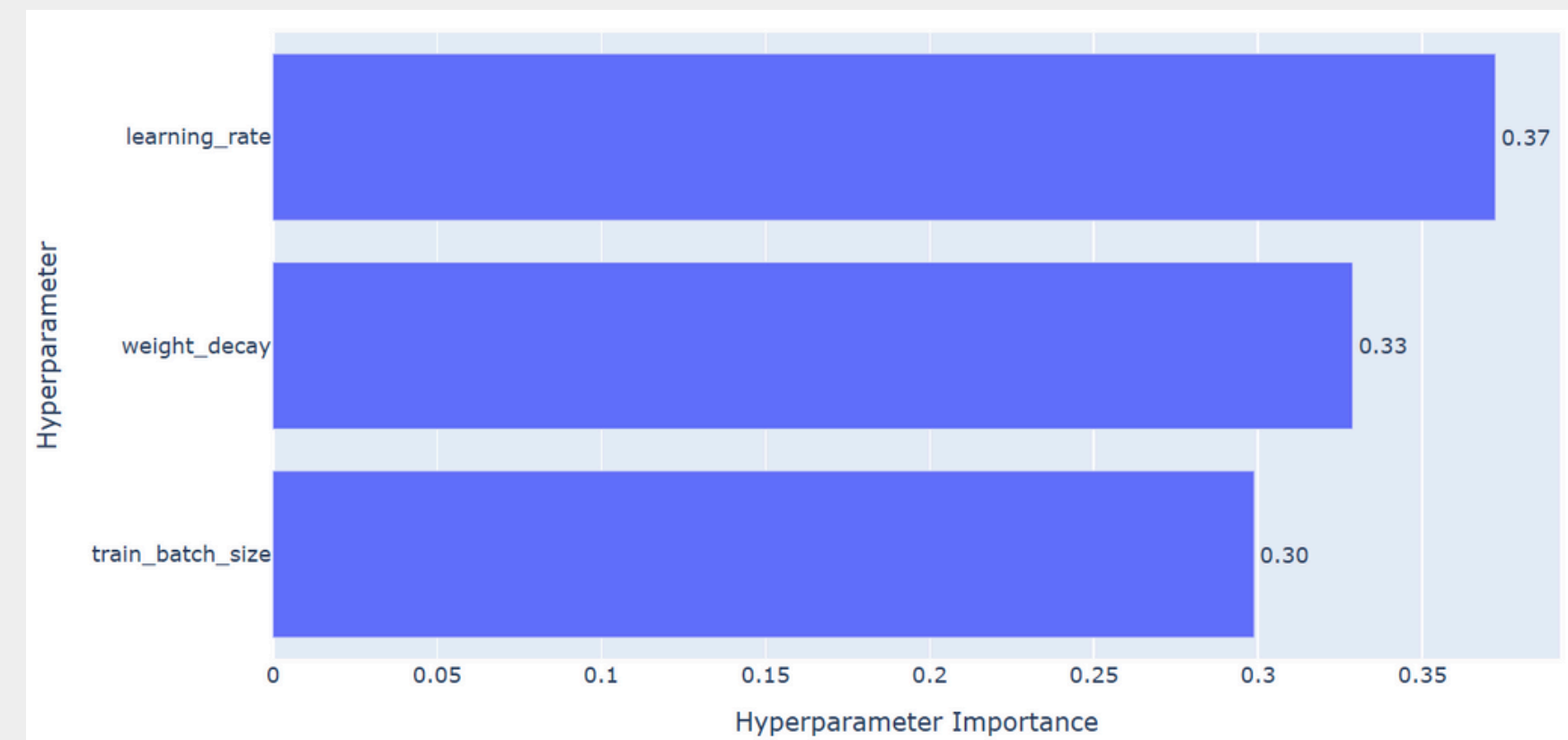


OPTUNA

HYPERPARAMETER TUNING

Limited to a subset of 1000 samples and 3 trials to keep training time low.

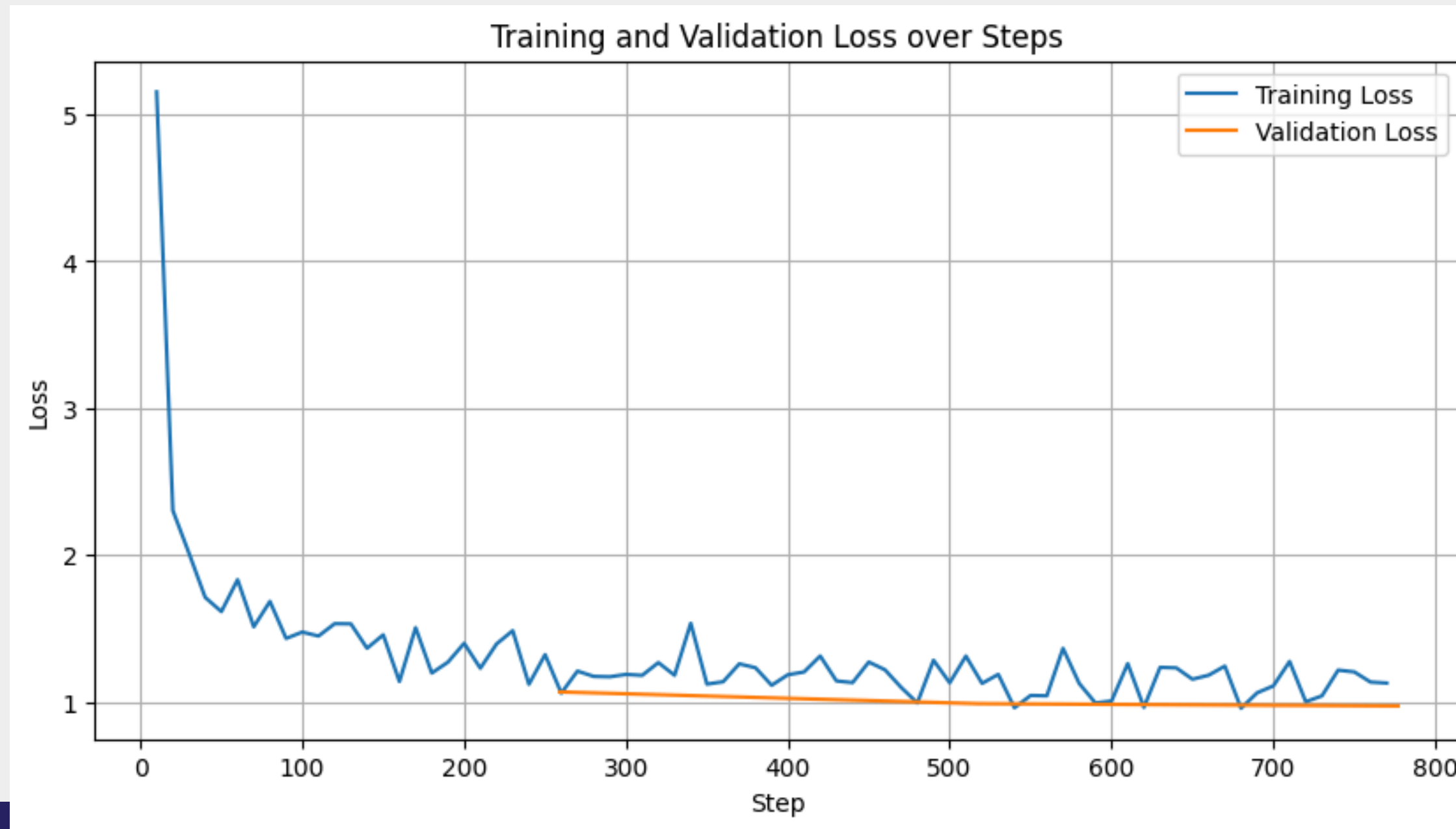
Learning Rate = 1.74e-4	Most influential for training stability and convergence speed
Max Input Length = 512	Needed to capture full meeting context without truncation
Batch Size = 8	Balanced between memory efficiency and training stability
Num Train Epochs = 3	Enough for the model to learn patterns without overfitting
Weight Decay = 0.0465	Helps with regularization and generalization



optuna.importance

Evaluates how much each hyperparameter contributes to the objective metric (ROUGE score)
This helped us prioritize the most influential parameters (e.g., learning rate, input length) during tuning

TRAINING AND VALIDATION LOSS



While training and validation loss may appear high, this is expected since loss is calculated at the token level, penalizing every mismatch. In contrast, ROUGE and METEOR evaluate summaries at the sequence level, focusing on meaningful phrase overlap – which better reflects real-world summarization quality.

Rouge Scores

ROUGE -1

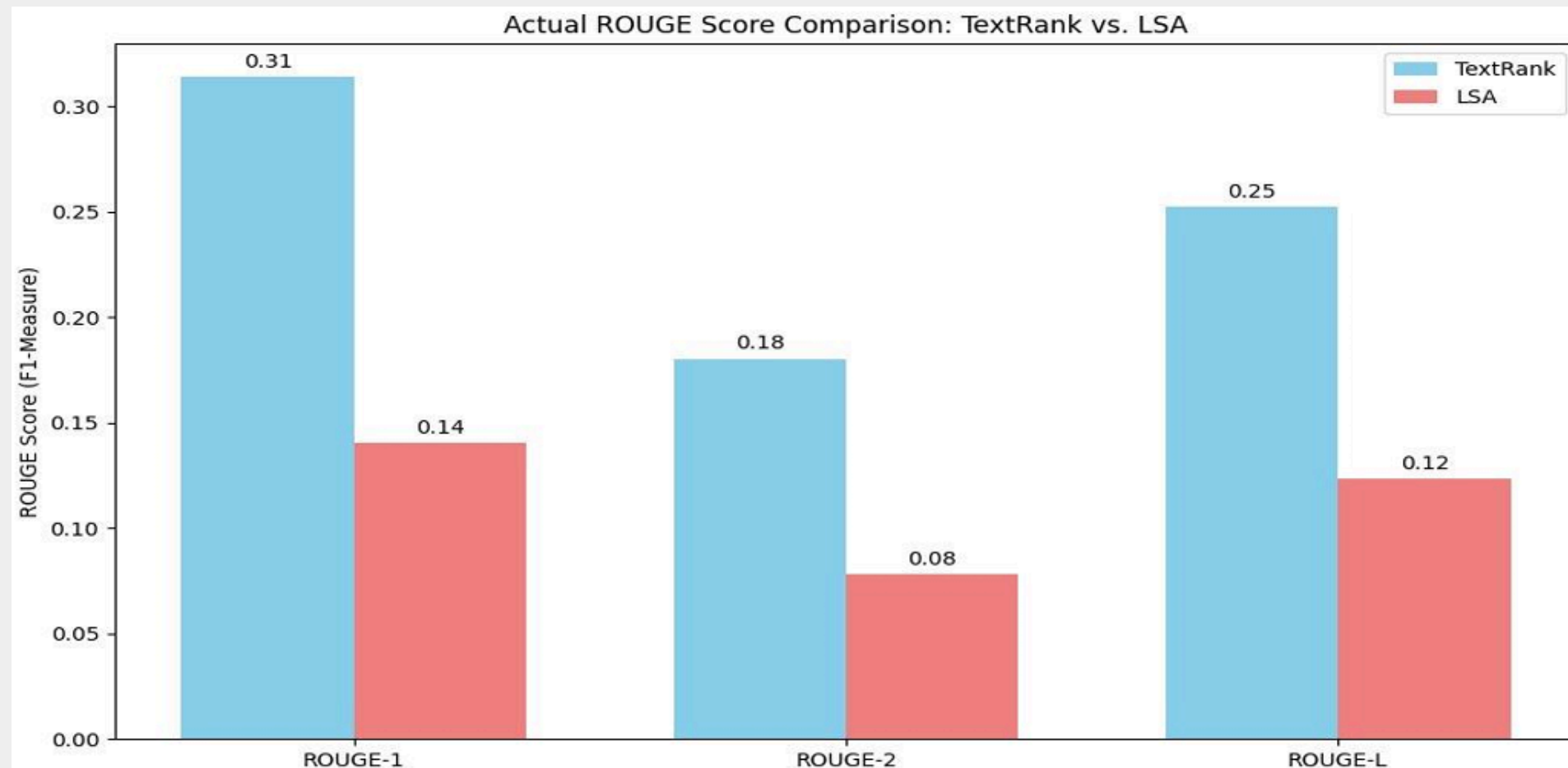
Overlap of words between generated & reference

ROUGE-2

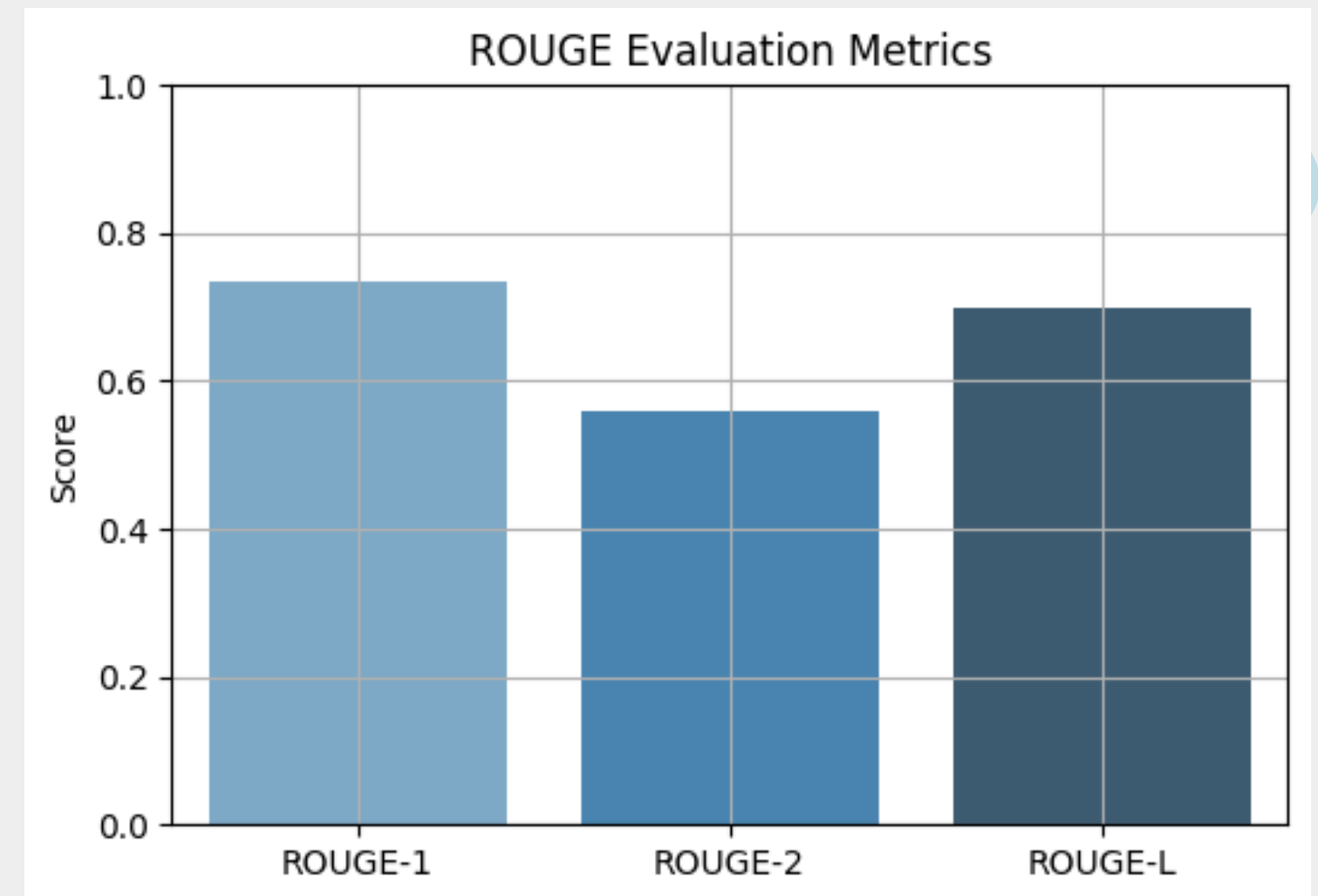
Overlap of words pairs between generated & reference

ROUGE-L

Longest common subsequence (captures fluency/overall structure/influence)



Unsupervised ML models



Text-to-text Transfer Transformer

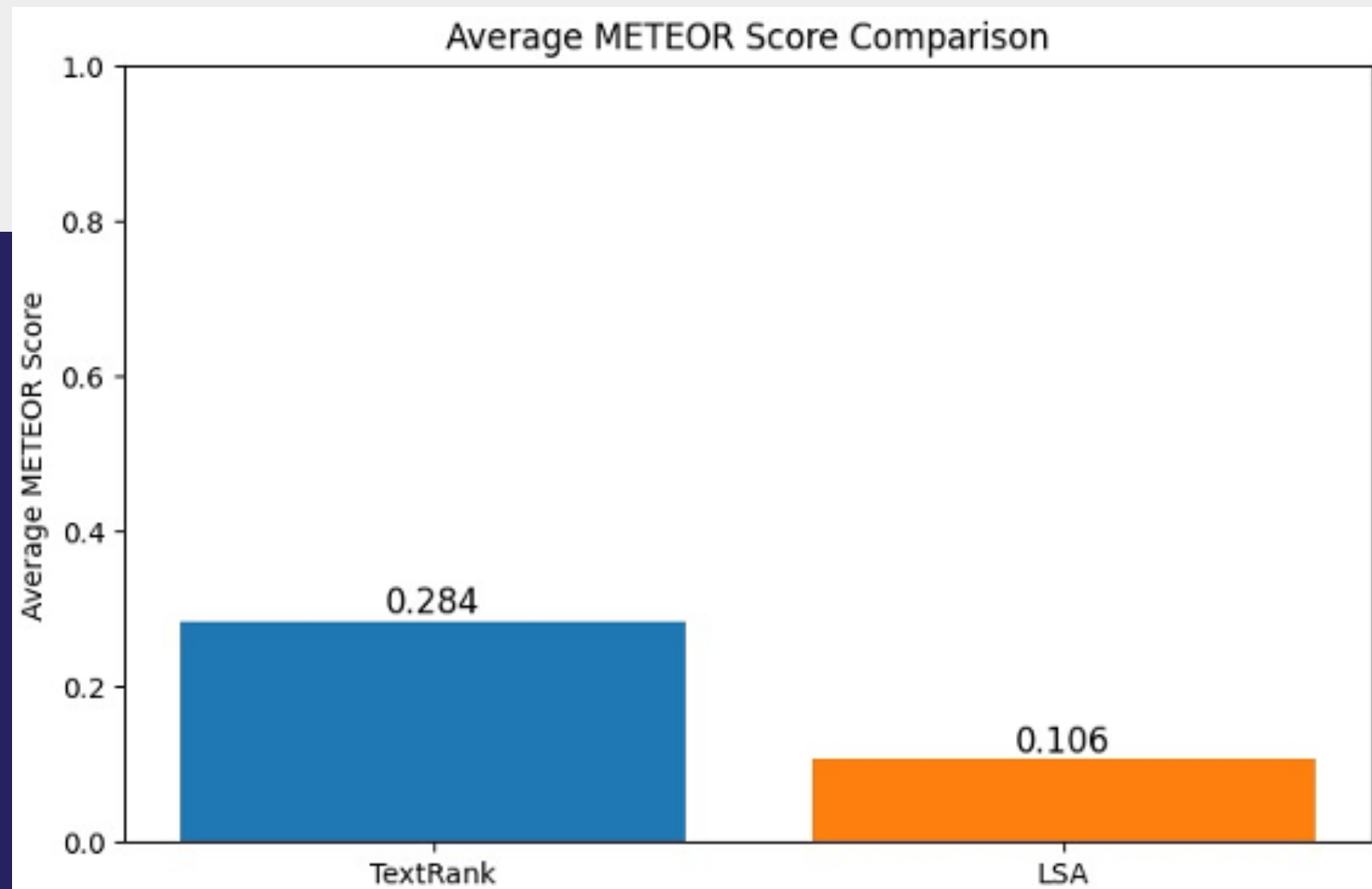
METEOR SCORES

METRIC FOR EVALUATION OF TRANSLATION OF EXPLICIT ORDERING

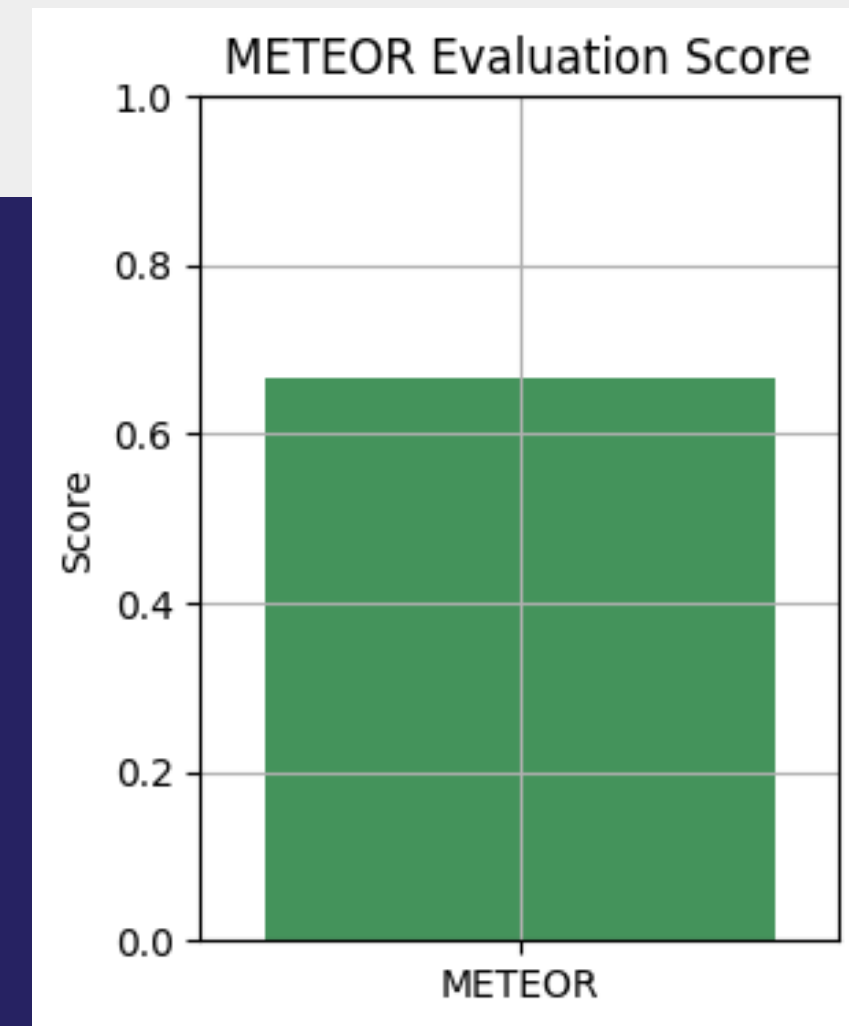
Evaluates how similar a generated summary is to a reference summary.

Considers precision, recall, and word order.

Scores above 0.4 are considered strong for abstractive summarization tasks like T5 on meeting transcripts.



Unsupervised ML Models



Text-to-Text Transfer Transformer

DEMO

127.0.0.1:7860

 **Smart Meeting Summarizer**

Efficiently generate bullet-point summaries, tone, and action items from any meeting transcript.

Choose a preset example (optional)

Weekly Check-in – Launch Delay

Summary Style

☐ Concise

☒ Detailed

Or Paste Your Meeting Transcript Below

Speaker 1: We're behind on the mobile app launch, UI integration is not done.

Speaker 2: Dev team estimates 4 more days of work, and QA will need another 3 days after that.

Speaker 3: We'll push the launch to next Wednesday. Notify Product and Sales teams.

Speaker 4: Understood. I'll update the release calendar.

 **Generate Summary**

 Bullet Summary

- The launch of the mobile app will be pushed to next Wednesday as the UI integration is not done yet
- Speaker 4 will update the release calendar and notify the product and sales teams that the launch will be delayed until next Wednesday
- The meeting was held by the developer team

 Tone of the Meeting

Positive

 Key Action Items

- Dev team estimates 4 more days of work, and QA will need another 3 days after that

 **Download Summary as CSV**

 Download File

CLICK HERE FOR DEMO



Smart Meeting Summarizer

Efficiently generate bullet-point summaries, tone, and action items from any meeting transcript.

Choose a preset example (optional)

Team Sync – Q4 Rollout



Summary Style



Concise



Detailed

Or Paste Your Meeting Transcript Below

Speaker 1: The Q4 feature rollout is scheduled for October 15, but backend progress is only at 60%.
Speaker 2: QA testing starts on September 28, and we plan to hold the first internal demo on October 1.
Speaker 3: DevOps will fix the staging deployment issue by Wednesday.
Speaker 4: Marketing needs user retention numbers and SLA targets by September 25 for the press release.
Speaker 1: Let's finalize all metrics before the weekend.
Speaker 2: Great progress so far. Thanks, everyone!



Generate Summary



Bullet Summary

- The Q4 feature rollout is scheduled for October 15
- Backend progress is only at 60%
- QA testing starts on September 28 and the first internal demo is planned for October 1
- Marketing needs user retention numbers and SLA targets by September 25 for the press release



Tone of the Meeting

Neutral



Key Action Items

- The Q4 feature rollout is scheduled for October 15, but backend progress is only at 60%
- DevOps will fix the staging deployment issue by Wednesday
- Let's finalize all metrics before the weekend



Download Summary as CSV



Download File



The background features several light blue circles of varying sizes scattered across the white space. A solid dark blue horizontal bar is positioned at the bottom of the image.

THANK YOU